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Offline video-based photogrammetry for EEG electrode localization

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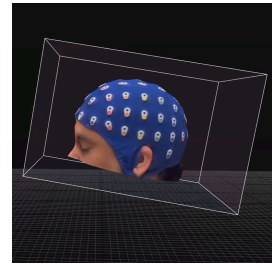
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We use this protocol and it's working

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Disclaimer

The authors do not take responsibility for the accuracy or reliability of user-generated data. Individuals using this protocol should receive appropriate training and establish inter-rater reliability before implementation. Users are responsible for obtaining all required ethics approvals and informed consent, and for handling identifiable participant data in accordance with applicable institutional policies and local regulations.

Abstract

This protocol describes an offline video-based photogrammetry workflow to generate 3D head models for EEG electrode localization from video recorded on a smartphone, tablet, digital camera, or other handheld device. The frames extracted from the video are reconstructed locally using RealityCapture, providing a low-cost, privacy-preserving alternative to cloud-based photogrammetry applications that require uploading identifiable participant data to external servers. Following this protocol produces a textured 3D head model suitable for electrode digitization in software such as Brainstorm, EEGLAB, FieldTrip, or MeshLab. The resulting electrode coordinates can then be scaled to anatomical units for use in EEG source localization and related neuroimaging analyses.

Image Attribution

Unless otherwise indicated, figures and screenshots were created by the author. Screenshots depicting the *RealityCapture* software interface are reproduced for instructional purposes. Selected screenshots were enhanced using Google's Gemini solely to improve image resolution and readability. No scientific or informational content was added, removed, or otherwise altered. The example participant shown throughout this protocol is the author (JPD), who has provided consent for the use and publication of these images.

Guidelines

Purpose: This protocol provides step-by-step instructions for acquiring video recordings and generating 3D head models for EEG electrode localization using an accessible, low-cost, offline photogrammetry workflow.

Privacy Considerations: Use a handheld video recording device, such as a digital camera, smartphone, or tablet. Whenever possible, use a device without an active cellular or Wi-Fi connection to minimize the risk of unintended data transmission. If desired, participants may wear a face mask to further protect their identity, provided it does not obscure any electrodes or fiducial landmarks.

Materials

Recording equipment

- Video camera (e.g., a digital camera, smartphone or tablet)
- Soft tape measure

Participant setup

- EEG cap with clearly labeled electrodes
- Stable, non-swivel chair with back support
- Diffuse light source
- *Optional:* hair clips or ties, surgical-style face mask for anonymity

Computer and software

- Windows PC with a CUDA-capable NVIDIA GPU (≥ 8 GB VRAM recommended)
- ≥ 8 GB system RAM
- RealityCapture (Epic Games), version 1.5 or later

Before start

Environment and Room Setup

- Conduct video acquisition in a room with sufficient, evenly distributed lighting to ensure clear visualization of the EEG cap, electrodes, and fiducial landmarks.
- Remove or cover reflective surfaces, including mirrors, glass panels, and metallic objects, as reflections can introduce errors during image alignment and 3D reconstruction.
- Provide adequate space around the participant (approximately 1 to 2 meters of clearance) to allow the camera operator to move freely and capture all viewing angles.
- Seat the participant in a stable, non-swivel chair with back support to help maintain a consistent posture throughout recording.



Participant Setup

5m

1 Prepare Participant for Video Recording

- 1.1 Have the participant take a seat in a chair placed in the center of the room.
- 1.2 Measure and record the participant's head circumference using a flexible measuring tape. Position the tape so that it passes just above the eyebrows at the front of the head, above the ears on both sides, and around the most prominent part of the back of the head (inion). Ensure that the tape remains level and is snug against the scalp without compressing the skin. Record the head circumference in centimeters, to the nearest 0.1 cm when possible. 1m 
- 1.3 Place the EEG cap on the participant's head, fitted per manufacturer's instructions. 4m

Note

For EEG caps with manually inserted electrodes, perform the video recording before electrode placement to prevent the electrode wires from obstructing the recording.

2 Ensure Visibility of Electrodes and Fiducials

- 2.1 Confirm that all electrodes and their labels are visible and unobstructed.
- 2.2 Ensure fiducial landmarks (e.g., nasion, inion, preauricular points) are clearly visible and not obscured by hair or wires.
- 2.3 If necessary, gently move hair away from these areas and secure it underneath the EEG cap or with accessories like clips or hair ties.

Video Recording

1m 30s

3 Record Video of Participant Wearing EEG Cap

1m



- 3.1 Instruct the participant to remain as still as possible throughout the recording.
- 3.2 Using a handheld device (e.g., a digital camera, smartphone or tablet), record approximately one minute of video.
- 3.3 While walking around the participant, ensure the video captures *all angles* of the EEG cap, including the sides, back, and *top of the head*.
- 3.4 Keep the participant's head and EEG cap centered in the video frame while including some background context to aid in the reconstruction process. Avoid cropping out the head or cap at any point during the recording.
- 3.5 Check video recording quality, and repeat if necessary.

Note

Ensure the recording environment remains unchanged during acquisition; moving objects, changing lighting conditions, or other environmental disturbances may negatively impact reconstruction quality.

4 Transfer Video to Windows PC

- 4.1 Using a USB cable, transfer the recorded video to the Windows PC that will be used for local 3D model reconstruction.

30s

3D Model Reconstruction

10m 30s

5 Prepare Software

Launch the *RealityCapture* software (v1.5.1).

30s

Software

RealityCapture

NAME

Windows 10 or 11

OS

Epic Games

DEVELOPER

6 Convert Video to Image Frames

Import frames extracted automatically from a video.

1m

6.1 In the **Workflow** tab, click on **Video Sequence**, and select the participant's video recording.

6.2 Allow the software to automatically split the video into image frames using a **Jump Length** of **0.3 seconds** (see Figure 1).

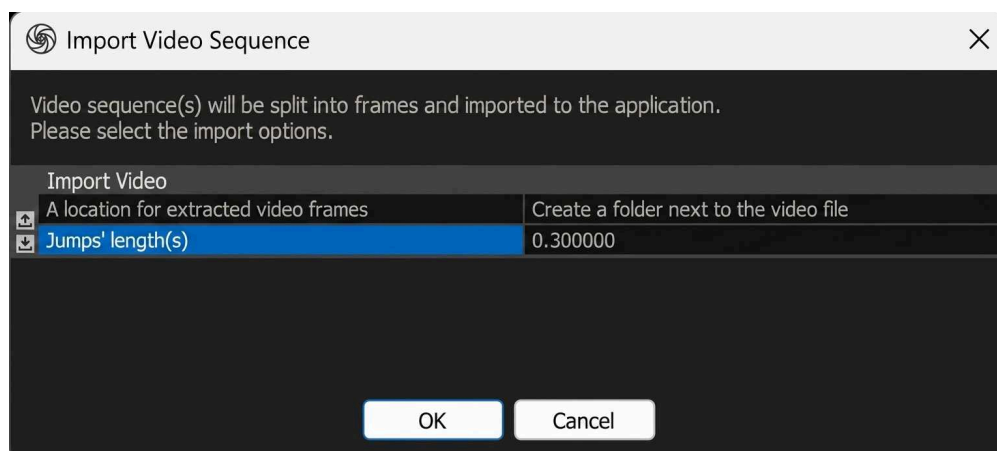


Figure 1. Example **Import Video Sequence** dialog showing the settings used to extract image frames from the video recording. The extracted frames are automatically saved to an output folder in the same directory as the original video file.

Note

A one-minute video recorded at 30 fps can produce up to 1,800 frames. Using a *Jump Length* of 0.3 seconds extracts approximately 200 images for reconstruction, which is sufficient to generate an accurate 3D model.

7 Align Images

Register all input images to estimate their camera positions and generate a sparse 3D point cloud.

- 7.1 Navigate to the **Alignment** tab, and select **Align Images**. Upon completion, a sparse 3D point cloud reconstruction of the entire scene will appear (see Figure 2).

1m

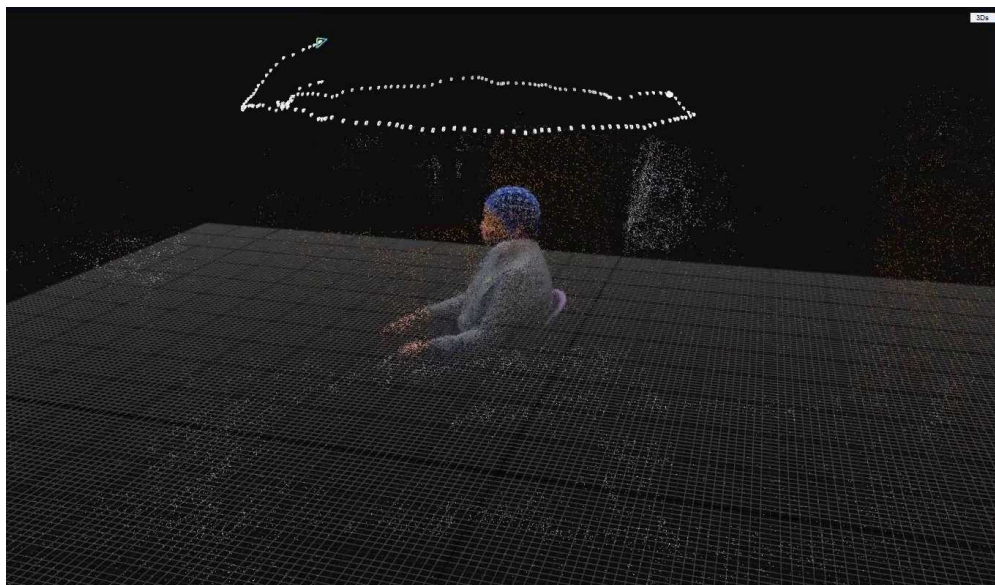


Figure 2. Example of a sparse 3D point cloud reconstruction. White dots above the point cloud represent the camera positions corresponding to each image frame.



Note

Poor 3D Point Cloud Reconstruction

After image alignment, review the number of **Components** and **Aligned Cameras** in the side panel. Ideally, all images should be assigned to a single component. Point Clouds that are broken up into multiple components typically indicate poor image alignment caused by insufficient image overlap, motion blur, or incomplete scene coverage. If multiple components are generated or other alignment issues are observed, use the following steps to identify and exclude low-quality frames before rerunning the alignment:

Exclude Low-Quality Frames

Review the input images after alignment and exclude blurry or poorly aligned frames.

1. **Identify images with fewer points.** In the side panel, review the Inputs and identify any frames with substantially fewer points than the remaining frames. Click on these frames and inspect the corresponding images for motion blur, poor image quality, or inadequate scene coverage.
2. **Review the camera poses.** Review the **Components** and their **Camera Poses**, and identify any frames with substantially fewer tie points than the surrounding frames.
3. **Exclude low-quality frames.** Hover over the selected image names, then click the **x** button that appears to exclude the images from the alignment.
4. **Rerun image alignment.** Rerun Image Alignment and review the resulting Components and Aligned Cameras to determine whether the alignment has improved. [➡ go to step #7](#)

8 Set Ground Plane

Set the orientation of the Ground Plane if the reconstructed point cloud appears in the wrong orientation (e.g., tilted, rotated, or upside down).

8.1 Under **Scene 3D Tools**, select **Set Ground Plane**.

8.2 Adjust the position of the point cloud relative to the ground plane to restore the correct orientation.

1m

9 Set Reconstruction Region

Restrict the scene reconstruction to a specific volume.

9.1 Navigate to the **Mesh Model** tab and select **Set Reconstruction Region**.

9.2 Adjust the reconstruction **bounding box** to include only the head and EEG cap (Figure 3). Click and drag the bounding box handles to resize the box, and drag the box itself as

1m

needed to reposition it around the head. Check the box from multiple viewing angles and continue adjusting its dimensions until the entire head and EEG cap are enclosed while excluding background elements such as the room, chair, torso, and shoulders.

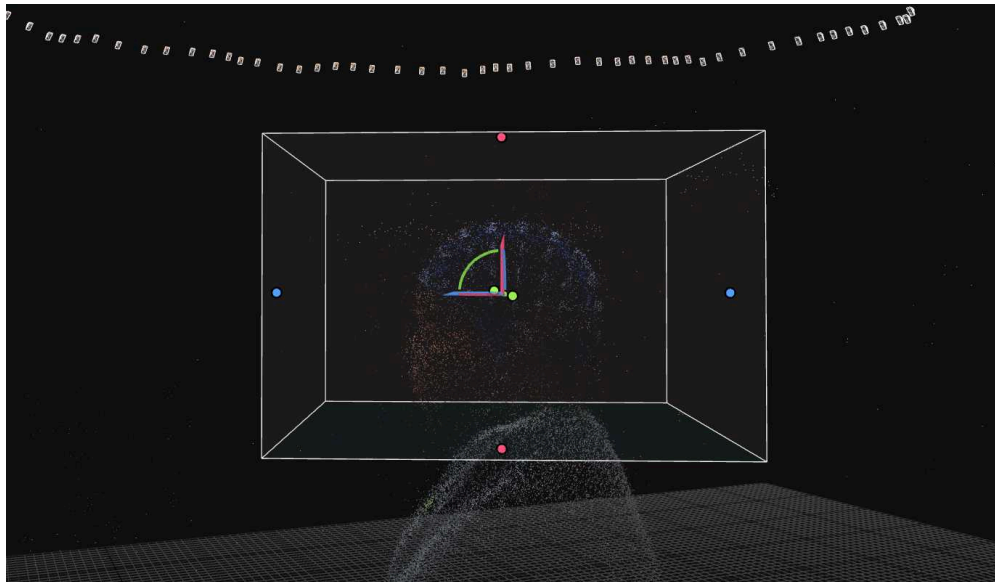


Figure 3. Example bounding box used to define the region of interest and exclude extraneous points from the 3D reconstruction.

10 Generate Mesh

Create a 3D model for the currently selected component.

- 10.1 In the **Mesh Model** tab, under **Create Model**, select **Normal Detail** to generate the 3D model (see Figure 4).

1m

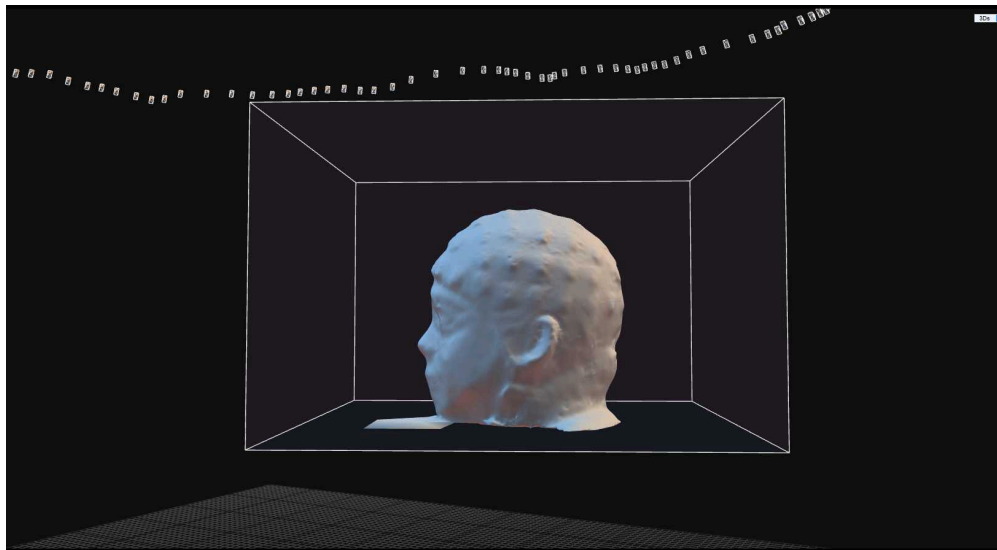
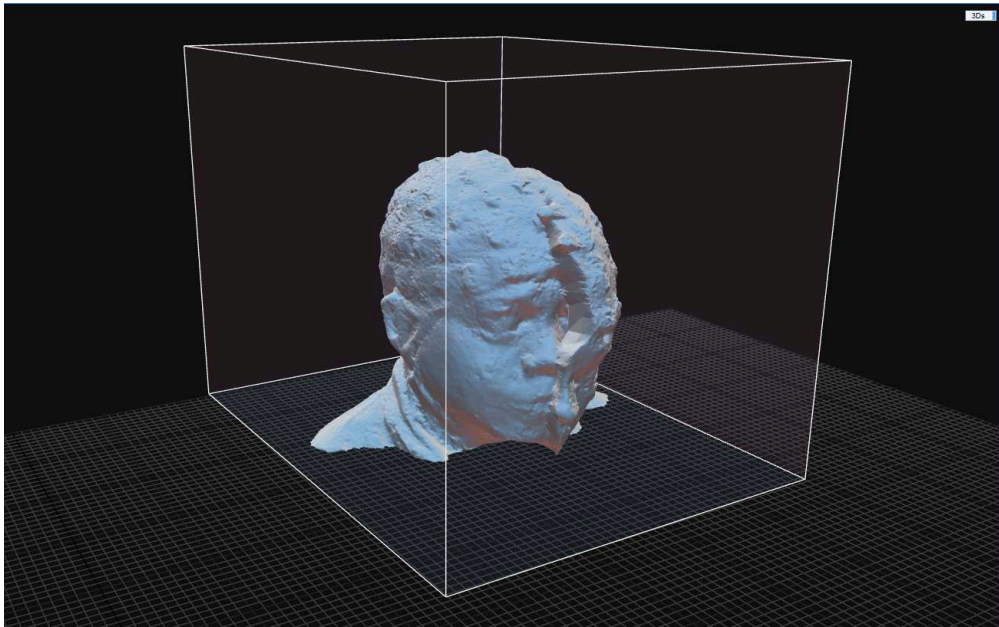


Figure 4. Example 3D mesh generated from the aligned images using the **Normal Detail** reconstruction setting.

Note

Misaligned 3D Mesh

After mesh generation, inspect the reconstructed 3D model for regions that appear duplicated, distorted, or incorrectly aligned. Misaligned portions of the mesh may result from errors in image alignment, particularly when similar features are incorrectly matched across images or when there is insufficient image overlap. If portions of the reconstructed model are misaligned, use control points to improve the image alignment and regenerate the mesh.



Example of a misaligned 3D mesh reconstruction. Misalignment typically appears as duplicated or distorted regions of the head resulting from reconstruction errors caused by participant movement, insufficient image overlap, motion blur, or other image alignment failures.

Correct Misalignment Using Control Points

Use control points to identify the same physical location across multiple images to establish correspondences between images.

1. **Identify the misaligned region.** Inspect the reconstructed model and determine which portion of the model is incorrectly aligned. Review the camera images associated with both the correctly aligned and misaligned portions of the reconstruction.
2. **Create a control point.** Open an image containing a clearly identifiable feature, such as an electrode, fiducial marker, or other distinct anatomical or physical landmark on the participant's head. Add a control point at the center of the selected feature.
3. **Locate the same feature in additional images.** Identify the same feature in several images captured from different viewing angles and place the corresponding control point as precisely as possible at the same physical location in each image. When correcting misalignment, include images associated with both the correctly aligned and misaligned portions of the model.

4. **Rerun image alignment and mesh generation.** After adding the control points, rerun **Steps 8 to 11**. The software will incorporate the manually defined correspondences when estimating the camera poses. [➡ go to step #7](#)
5. **Inspect the updated reconstruction.** Confirm that the previously misaligned regions are correctly aligned and that the control points correspond to the same physical locations across images. If misalignment persists, add additional control points or review existing control points for inaccurate placement before rerunning alignment.

11 Cut Model By Box

Remove any remaining unnecessary parts of the model to further anonymize it and reduce its file size.

11.1 In the **Scene 3D Tools**, under **Mesh Model**, select **Cut by Box**.

11.2 Adjust the bounding box to remove unnecessary facial features (see Figure 5). Ensure the following anatomical landmarks remain fully visible:

- *Nasion* (depression at the top of the nose, between the eyes)
- *Inion* (external occipital protuberance at the back of the head)
- *Left preauricular point* (depression immediately anterior to the left tragus)
- *Right preauricular point* (depression immediately anterior to the right tragus)
- *Vertex* (top of the head, approximately at the Cz electrode location)
- *All EEG electrodes*

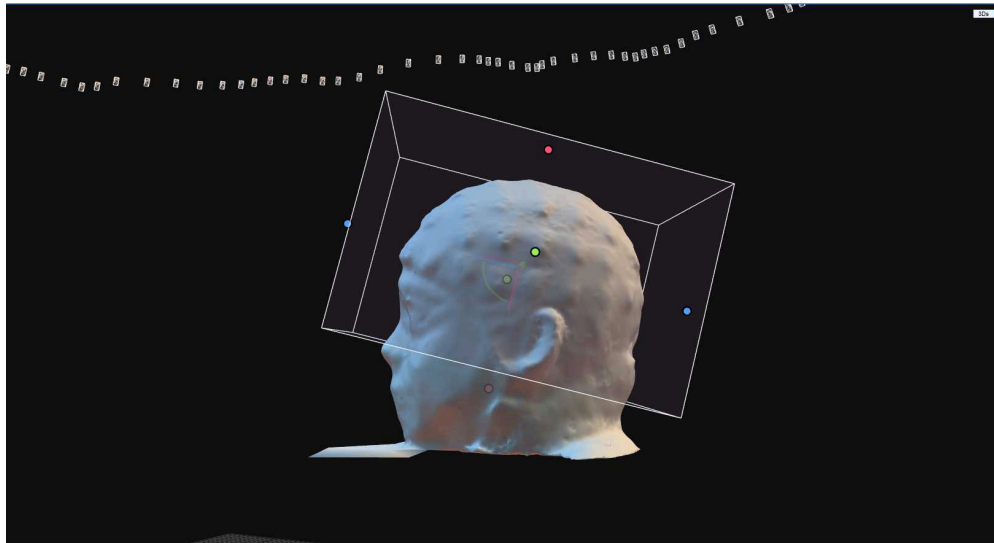


Figure 5. Example bounding box used for **Cut by Box** to remove unnecessary portions of the reconstructed mesh while preserving anatomical landmarks and EEG electrodes.

- 11.3 Select **Cut outer** (see Figure 6) to remove the mesh outside the bounding box.

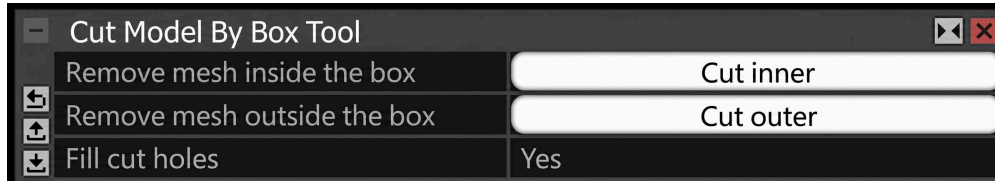


Figure 6. Cut by Box dialog used to remove mesh outside or inside the selected bounding box.

- 11.4 Inspect the cropped model (Figure 7) and verify that unnecessary portions of the mesh have been removed while the EEG cap, electrodes, and required anatomical landmarks remain intact.

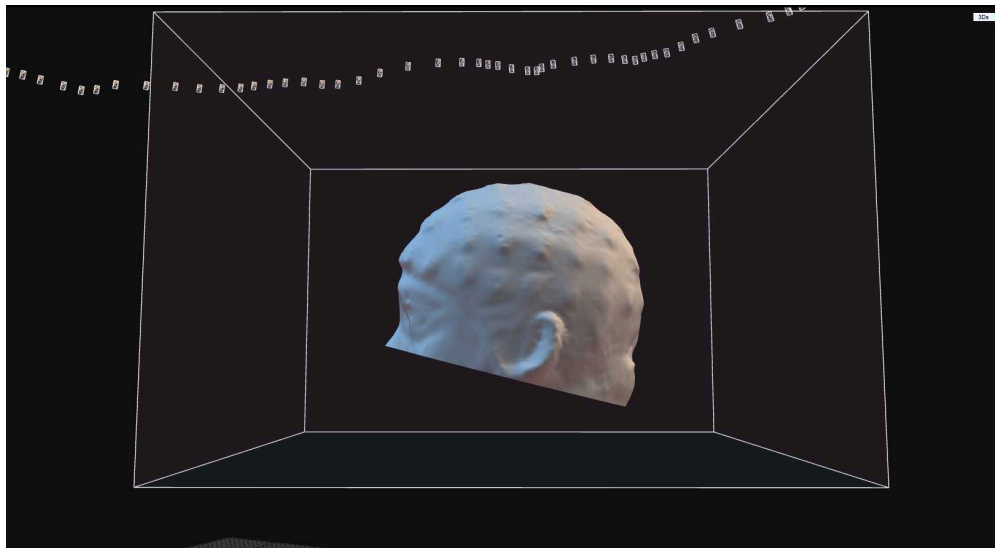


Figure 7. Example 3D mesh after applying **Cut by Box**. The cropped model should retain the EEG cap, electrodes, and required anatomical landmarks.

12 Smooth Mesh Model

Smooth down rough surfaces of the model.

- 12.1 In **Scene 1D Tools**, select **Smoothing Tool**, then click **Smooth** (see Figure 8).

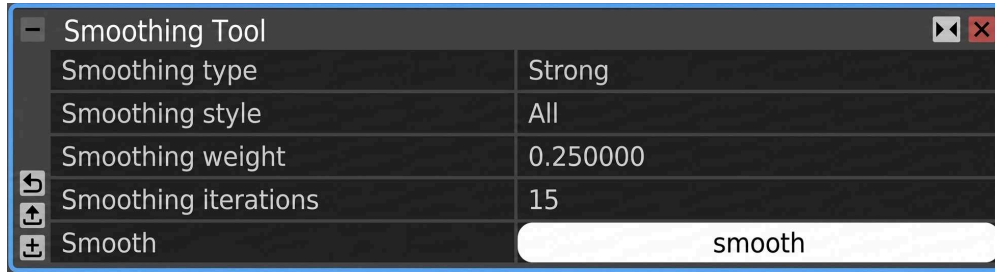


Figure 8. Example **Smoothing Tool** settings applied to the reconstructed mesh.

13 Generate Texture

Generate the model texture and apply color to the 3D model.

- 13.1 In the **Mesh Model** tab, select **Texture** to generate a fully textured, colored 3D head model (see Figure 9).

1m



Figure 9. Fully textured 3D head model after texture generation.

14 Export Model

Export the model as a mesh and texture file.

- 14.1 In the **Mesh Model** tab, under **Export**, select **Dense Mesh Model** to export the mesh.
- 14.2 Specify the file name, then save the model in **Wavefront obj (*.obj)** format.

- 14.3 Under **Texturing Settings** (see Figure 10), set **Export to a Single Texture File** to **Yes**, then save the model.

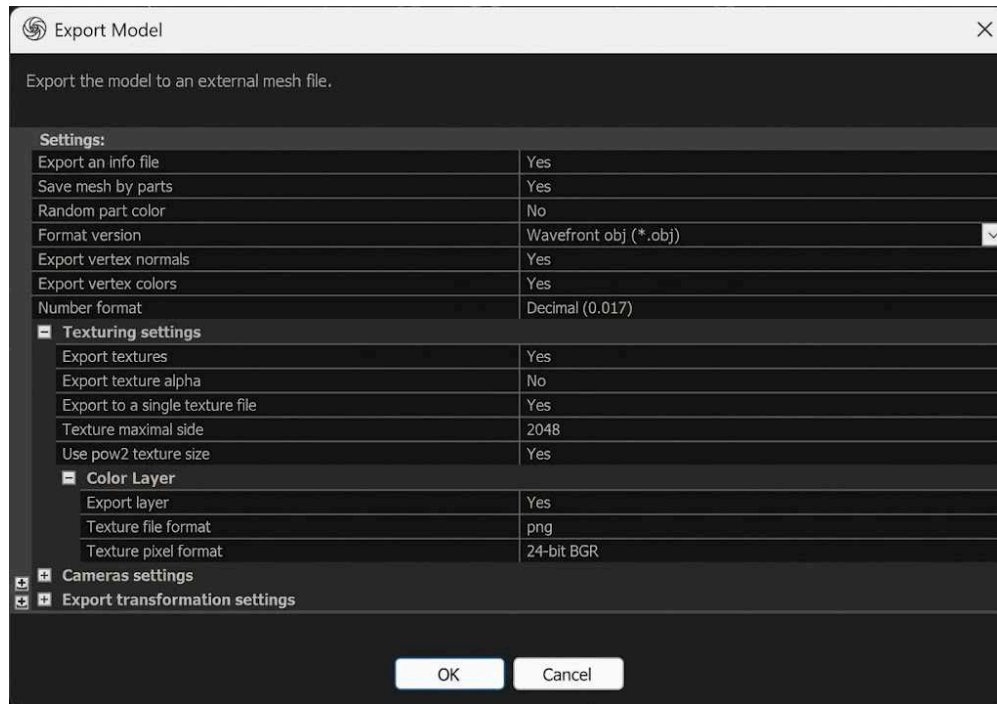


Figure 10. Example export settings for saving the reconstructed model as a Wavefront (.obj) mesh with a single texture file.

Note

A successful export produces three files: an **.obj** mesh file, a **.png** texture image, and an **.mtl** material file.

Spatial Localization of Electrodes

30m

- 15 After creating the 3D model, digitize the electrode locations using one of the following supported software packages:
- FieldTrip (Homölle & Oostenveld, 2019)
 - EEGLAB (*get_chanlocs* plugin; Lee & Makeig, 2024)
 - Brainstorm (Chinara et al., 2024)
 - MeshLab (Diaz-Fong et al., 2024)

30m

Coordinate Normalization

- 16** The mesh produced by *RealityCapture* is reconstructed at an arbitrary scale because photogrammetry estimates 3D geometry from the relative positions of features across overlapping 2D images but, without a known physical reference, cannot determine their absolute dimensions. To rescale electrode coordinates into true anatomical units (cm or mm), use the head circumference recorded in **Step 1** as a ground truth measurement.
- 16.1** Identify the electrodes located along the perimeter of the EEG cap that define the head circumference.
- 16.2** Fit a plane to those points by least-squares (via SVD) and rotate them so the plane is parallel to XY.
- 16.3** Fit an ellipse to the in-plane (2D) coordinates by algebraic least-squares.
- 16.4** Compute the ellipse perimeter P from its semi-axes a and b using a series expansion.
- 16.5** Compute the scale factor s by dividing the measured head circumference from Step 1 by the computed ellipse perimeter.
- $$s = C_{\text{measured}} / P$$
- 16.6** Apply the scale factor to all original electrode coordinates.

Note

Scaling to true anatomical dimensions could also be applied directly to the 3D mesh model. For example, a scaled reconstruction of the participant's head could potentially be used to warp a template MRI to the participant-specific head shape to generate a pseudo-MRI (Jaiswal, Nenonen, & Parkkonen, 2025).



Protocol references

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